Role Adherence: A Metric for Evaluating Role Consistency in Multi-Turn Conversational AI

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Abstract

As conversational AI systems are increasingly deployed in role-constrained settings — customer support agents, educational tutors, medical assistants — the ability to measure whether a model consistently adheres to its assigned role becomes a core quality requirement. This paper proposes **Role Adherence**, a metric for the Gaussia evaluation framework that quantifies the degree to which an AI assistant maintains its defined role throughout a multi-turn conversation. We define adherence constructively, describe a dual-path scoring strategy conditioned on the availability of ground-truth data, and discuss design decisions and alternative formulations considered during development. No empirical validation is presented; this paper constitutes a design proposal intended to guide the implementation of the metric within the Gaussia SDK.

1 Introduction

Modern LLM-based assistants are typically deployed with a system prompt that defines their role: the topics they should cover, the tone they should adopt, the behaviours they should exhibit, and the boundaries they should respect. Despite this, models frequently deviate from their assigned roles — answering out-of-scope questions, abandoning their defined persona, or failing to exhibit the behaviour their role requires.

Existing evaluation frameworks either measure role adherence at the level of individual turns in isolation [Zheng et al., 2023], or rely entirely on LLM-based judges, which introduce reproducibility and cost concerns [Deriu et al., 2021]. This paper proposes a more complete metric that evaluates adherence across the full conversational context, supports both structured and unstructured role definitions, and offers a deterministic scoring path when reference data is available.

2 Problem Statement

Given a multi-turn conversation between a user and an AI assistant, and a definition of the role the assistant is expected to play, we want to compute a scalar score representing how consistently the assistant adhered to that role throughout the conversation.

Let a conversational session be a sequence of assistant turns $t_1, t_2, \ldots, t_n$, and let R denote the role definition. The Role Adherence score is defined as:

$$\text{RoleAdherence}(R, T) = \frac{1}{n} \sum_{i=1}^{n} \text{adhere}(t_i,\ T_{<i},\ R) \tag{1}$$

where $T_{<i}$ is the conversation history up to turn i, and adhere$(\cdot) \in [0, 1]$ is a per-turn adherence score. The session score lies in $[0, 1]$, where 1 indicates full adherence across all turns. The range of adhere$(\cdot)$ and how it is computed depend on the scoring path selected (Section 5).

3 Defining Adherence

3.1 Constructive vs. Restrictive Adherence

Two competing definitions of adherence are possible.

Restrictive adherence asks: *did the assistant avoid going outside the scope of its role?* A turn is compliant if it does not address topics or exhibit behaviours explicitly excluded by the role. This definition only penalises negative deviations.

Constructive adherence asks: *did the assistant actively behave as its role requires?* A turn is compliant only if it exhibits the behaviour, tone, and knowledge a properly role-playing assistant should exhibit. This definition penalises both negative deviations (scope violations) and positive failures (failing to act as the role demands).

Consider an assistant assigned the role of a technical support agent. If the user asks a question within the assistant's domain and the assistant responds "I don't know", restrictive adherence would record no violation — no out-of-scope content was produced. Constructive adherence would record a failure, because a role-adherent support agent is expected to answer questions in its domain.

We adopt the **constructive definition** as the primary formulation for this metric, as it provides a more meaningful signal for production quality assessment. Restrictive adherence remains a valid and simpler formulation for use cases where the primary concern is containment rather than quality of execution. Both definitions are implementable within the proposed schema.

3.2 Conversational Context

When evaluating whether turn t_i adheres to role R, the evaluator receives the full conversation history $T_{<i} = \{t_1, \ldots, t_{i-1}\}$. A turn evaluated in isolation may appear off-role but be entirely appropriate given prior context. Conversely, a turn evaluated in isolation may appear neutral but represent a pattern of escalating deviation only visible across the conversational arc.

The alternative — evaluating each turn independently — is computationally cheaper and avoids context-window constraints in very long conversations. We note this as a valid trade-off for resource-constrained deployments but do not adopt it as the default behaviour.

4 Role Definition Schema

4.1 Placement in the Data Model

The role definition is a property of the conversational session, not of the metric. Two sessions with the same conversation content but different assigned roles represent fundamentally different evaluation scenarios. Accordingly, `chatbot_role` is defined as a field on the `Dataset` object.

An alternative considered was to pass the role as a parameter at metric instantiation time, which would allow evaluating the same dataset against multiple roles. This was discarded on the grounds that a dataset already carries implicit contextual assumptions: the way a user interacts with a customer support agent differs from how they interact with a coding assistant, even if the raw conversation text were identical. Separating the role from the session would break this coupling and weaken the semantic integrity of the dataset object.

4.2 Format: String or Structured

Role definitions in practice range from a single prose sentence to detailed specifications with topic lists, tone requirements, and explicit restrictions. To accommodate both, we propose that `chatbot_role` support two formats:

String format — free-form natural language description of the role. Flexible and low-friction for exploratory evaluation.

Structured format — a `RoleDefinition` object with explicit fields: `description` (required), `allowed_topics` (optional list), `tone` (optional string), and `restrictions` (optional list). This format reduces ambiguity in judge prompts and unlocks constraint-based deterministic evaluation (Section 5.3.4).

Both formats are valid inputs. The evaluation pipeline adapts accordingly. The structured format is proposed as the preferred option for production deployments where reproducibility and auditability are priorities.

5 Scoring Methodology

5.1 Dual-Path Strategy

The scoring strategy bifurcates based on the availability of ground-truth data:

- **Without ground truth**: an LLM-as-a-judge evaluates each assistant turn against the role definition and conversation history.
- **With ground truth**: a deterministic metric compares each assistant turn against an expected role-adherent reference response.

This separation is principled. LLM-based judges are well-suited for open-ended evaluation where no reference answer exists [Zheng et al., 2023], but introduce non-determinism, model dependency, and inference cost. When reference data is available, deterministic metrics offer reproducibility, auditability, and lower computational cost.

5.2 LLM-as-a-Judge Path

When no ground truth is available, each assistant turn t_i is evaluated by an LLM judge that receives: (1) the role definition R; (2) the conversation history $T_{<i}$; and (3) the turn to evaluate t_i. The judge prompt is constructed to implement the constructive adherence definition: the judge is asked not only whether the turn avoids scope violations, but whether it actively reflects the expected behaviour of the defined role.

The judge can be configured to return either a **continuous score** in $[0, 1]$ or a **binary score** in $\{0, 1\}$. Continuous scoring is more expressive — a response that mostly adheres but deviates in tone might receive 0.7 rather than a hard 0 — but LLMs are generally not well-calibrated for numeric outputs, meaning the same response may receive different scores across calls. Binary scoring trades nuance for consistency and reproducibility. We propose binary scoring as the default for the LLM judge path, with continuous scoring available as a configurable option for deployments that prioritise granularity over stability. When `include_reason` is enabled, the judge additionally returns a natural-language justification for each per-turn score.

5.3 Deterministic Path

When a `ground_truth_assistant` response is available for each turn, the following metrics are proposed as scoring functions. The per-turn score is continuous in $[0, 1]$ and is averaged across turns to produce the session score.

5.3.1 BERTScore

BERTScore [Zhang et al., 2020] computes semantic similarity between two texts by extracting contextual token embeddings from a pre-trained language model and computing precision, recall, and F1 over token-level cosine similarities. Unlike surface-level metrics, it captures paraphrastic equivalence. We adopt BERTScore F1 as the **primary deterministic metric**:

$$\text{BERTScore}_{F_1} = 2 \cdot \frac{P_{\text{BERT}} \cdot R_{\text{BERT}}}{P_{\text{BERT}} + R_{\text{BERT}}} \tag{2}$$

Role-adherent responses are not expected to reproduce the exact phrasing of a reference. BERTScore tolerates natural linguistic variation while remaining independent of a secondary judge model.

5.3.2 Semantic Cosine Similarity

Cosine similarity between sentence-level embeddings [Reimers and Gurevych, 2019] is a lighter-weight alternative. A sentence encoder maps each response to a dense vector; adherence is scored as the cosine of the angle between the candidate and reference vectors:

$$\text{sim}(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \, \|\mathbf{b}\|} \tag{3}$$

Cosine similarity operates at sentence granularity rather than token granularity, making it faster but less sensitive to fine-grained semantic differences. BERTScore and cosine similarity are compatible and non-exclusive — they measure related but distinct levels of semantic granularity. Where computational resources permit, BERTScore is preferred.

5.3.3 Natural Language Inference

NLI models [Bowman et al., 2015] classify the relationship between two texts as *entailment*, *neutral*, or *contradiction*. Applied to role adherence, a candidate response that contradicts what a ground-truth role-adherent response asserts is a strong signal of non-adherence. NLI is proposed as a complementary signal for detecting explicit violations rather than measuring overall similarity. A *contradiction* label serves as a hard penalty regardless of similarity scores.

5.3.4 Constraint-Based Evaluation

When the role is defined in structured format, adherence can be partially evaluated through explicit constraint verification without any model inference:

- **Topic scope**: whether the response discusses topics outside `allowed_topics`.
- **Restriction compliance**: whether the response violates any item in `restrictions`.
- **Tone consistency**: whether the detected tone matches the expected `tone` field.

These checks are deterministic, fully reproducible, and computationally trivial. They are proposed as a first-pass filter that complements BERTScore or cosine similarity rather than replacing them.

5.3.5 Lexical Metrics: Considered and Discarded

ROUGE [Lin, 2004] and BLEU [Papineni et al., 2002] measure n-gram overlap between candidate and reference texts. Both were considered and discarded as primary metrics for this use case.

Role-adherent responses are not expected to reproduce the exact phrasing of a reference — they are expected to exhibit the same role-consistent behaviour with natural linguistic variation. N-gram metrics penalise paraphrases and reward surface-level copying, neither of which is desirable here. ROUGE and BLEU are noted as valid lightweight baselines for benchmarking purposes only.

6 Configuration Parameters

The metric exposes four optional parameters that control evaluation behaviour without altering the underlying methodology.

`threshold` (default: 0.5) — the minimum session score required to consider the assistant as role-adherent. Scores at or above this value are marked as passing. The threshold applies to the final session scalar produced by Equation 1.

`strict_mode` (default: `False`) — when set to `True`, the session score is converted to a binary output: 1 if the score equals 1.0 (every turn adheres), 0 otherwise. This also overrides `threshold` to 1.0. Strict mode is intended for high-criticality deployments — such as medical or legal assistants — where partial adherence is not acceptable.

`include_reason` (default: `True`) — when enabled and the LLM-judge path is active, the judge returns a natural-language justification alongside each per-turn score. Justifications aid interpretability and are included in the metric output for inspection.

`model` — the LLM used as the judge in the no-ground-truth path. Follows the `Guardian` abstraction already present in the Gaussia framework, allowing any supported provider to be substituted without changes to the metric logic.

7 Implementation Considerations

The metric extends the `Gaussia` base class. The `batch()` method iterates over each assistant turn in a session, selects the LLM-judge or deterministic path based on the presence of `ground_truth_assistant`, and appends a per-turn score. The session score is the mean of per-turn scores (Equation 1).

The `chatbot_role` field is read from the `Dataset` object. When `chatbot_role` is a `RoleDefinition`, the structured fields are serialised into the judge prompt and, where applicable, used for constraint-based evaluation. When it is a plain string, it is passed directly.

Statistical mode compatibility (Frequentist and Bayesian) follows the existing Gaussia pattern. No changes to the statistical layer are required.

8 Design Decisions Summary

Table 1: Summary of design decisions and alternatives considered.

Dimension	Decision	Alternative Considered
`chatbot_role` location	Field on `Dataset`	Parameter of the metric
Adherence definition	Constructive	Restrictive
Role format	`str | RoleDefinition`	String-only
Judge context	Full conversation history up to turn	Per-turn isolation
Scoring without GT	LLM-as-a-judge	—
LLM judge output mode	Binary (default), continuous (configurable)	Continuous-only
Scoring with GT	BERTScore (primary), cosine similarity (lightweight), NLI (violation detection)	ROUGE, BLEU (discarded)
Score output	Session-level scalar in $[0, 1]$	—

9 Future Work

The current proposal scopes Role Adherence to text-only conversations. Extending the metric to **multimodal settings** is identified as the primary direction for future development.

9.1 Multimodal Role Adherence

Conversational AI systems are increasingly capable of processing and generating content across multiple modalities — images, audio, and video alongside text. A role-adherent response in a multimodal context must satisfy the same constructive definition proposed in Section 3: the assistant must actively behave as the role requires, not merely avoid out-of-scope output. This requirement extends naturally to non-text modalities.

9.1.1 Role Definition Extensions

The `RoleDefinition` schema would require a `modalities` field specifying which content types the role operates over (e.g., `["text", "image"]`, `["text", "audio"]`). The presence of this field makes modality compliance an explicit dimension of adherence: an assistant whose role requires generating images but responds with text only fails constructive adherence regardless of the textual content quality.

9.1.2 Scoring in Multimodal Settings

The dual-path strategy generalises to multimodal content, but the concrete scoring functions differ by modality:

LLM-as-a-judge path — when the conversation contains non-text content, the judge must be a vision-capable or otherwise multimodal model. The judge logic and prompt structure remain unchanged; only the model requirement is elevated.

Deterministic path — BERTScore and cosine similarity over text embeddings do not apply to image or audio content. For image modalities, we propose exploring CLIP-based similarity [Radford et al., 2021], which embeds images and text into a shared semantic space and allows cross-modal comparison. For mixed responses combining text and image, a weighted combination of text-based and image-based similarity scores would be required, with the weighting strategy to be determined empirically.

9.1.3 Data Schema

The `Batch` schema currently models `query` and `assistant` as plain text fields. Supporting multimodal content would require these fields to accept structured payloads that can carry text, image references, or audio alongside appropriate metadata. This is a schema-level change with no impact on the metric formula or scoring logic.

9.2 Additional Directions

- **Empirical validation**: comparison of scoring strategies (BERTScore, cosine similarity, NLI, LLM judge) against human annotations of role adherence across diverse role types and domains.
- **Structured role extraction**: automatic construction of a `RoleDefinition` from a free-form system prompt using an LLM, enabling constraint-based evaluation without manual annotation.
- **Turn-level reporting**: exposing per-turn adherence scores alongside the session aggregate to allow fine-grained diagnosis of where role deviation occurs.
- **Cross-lingual evaluation**: role adherence in multilingual assistants, where role definitions and responses may be in different languages.

10 Conclusion

We have proposed Role Adherence, a metric for the Gaussia evaluation framework that measures how consistently an AI assistant maintains its assigned role across a multi-turn conversation. The metric adopts a constructive definition of adherence, evaluates each turn in full conversational context, and offers a dual scoring path — LLM-as-a-judge when no reference data is available, and deterministic semantic metrics (BERTScore, cosine similarity, NLI) when ground truth is present. The role definition is modelled as a first-class field on the session dataset, supporting both free-form string and structured formats. This paper constitutes a design proposal scoped to text-only conversations; extension to multimodal settings is identified as the primary direction for future work.

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